

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (EC)

December 2012

EC 207 Control Systems

Date: 6.11.2012, Tuesday

Time: 1:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) What do you mean by servomechanism system? Explain one of the examples of it. [04]
(b) List out the difference between Block diagram and signal flow graph techniques. [03]
- Q - 2 (a) Explain Gear train system. [05]
(b) Find the Force to voltage analogy of the given system in **Figure 1**. [05]
(c) Examine the stability by Routh's criterion of given transfer function: [04]
$$s^8 + 5s^6 + 2s^4 + 3s^2 + 1 = 0$$

OR

- Q - 2 (a) What do you mean by analogous system? Is it required? [05]
(b) Find the Force to current analogy of the given system in **Figure 2**. [05]
(c) Examine the stability by Routh's criterion of given transfer function: [04]
$$s^6 + 2s^5 + 8s^4 + 12s^3 + 20s^2 + 16s + 16 = 0$$
- Q - 3 (a) Using Block diagram reduction technique evaluate a transfer function C/R of a system given in **Figure 3**. [07]
(b) Obtain transfer function C/R from the signal flow Graph shown in **Figure 4**. [07]
(Use Mason's gain formula)

OR

- Q - 3 (a) Using Block diagram reduction technique evaluate a transfer function C/R of a given system in **Figure 5**. [07]
(b) Obtain transfer function C/R from the signal flow Graph shown of system given in **Figure 6**. (Use Mason's gain formula) [07]

SECTION - II

- Q - 4 (a) Define the terms (i) Delay Time (ii) Rise Time (iii) Peak Time (iv) Peak Overshoot [04]
(b) Find the steady state error for various types of standard test inputs for a unity feedback system with [03]

$$G(s)H(s) = \frac{k}{s(s+5)(s+10)}$$

(i) $k = 10$ (ii) $k = 200$.

- Q - 5 (a) A feedback system is given by following transfer function, Draw the bode plot and comment on stability [10]

$$G(s)H(s) = \frac{100(s+4)}{s(s+0.5)(s+10)}$$

- (b) Determine the stability of the system whose characteristics equation is [04]
 $s^6 + 4s^5 + 3s^4 - 16s^3 - 64s^2 - 48s = 0$

OR

- Q - 5 (a) The open loop transfer function of system is [07]

$$G(s)H(s) = \frac{K}{s(s+1)(s+5)}$$

Using Nyquist criterion, determine whether the closed loop system is stable or not.

- (b) For the given control system, sketch the polar-plot. [07]

$$G(s)H(s) = \frac{100}{s(s+4)(s+2)(s+8)}$$

- Q - 6 (a) Sketch the Root Locus for [07]

$$G(s)H(s) = \frac{K}{s(s^2 + 2s + 2)}$$

- (b) A system is described in Figure 7, Determine its State equations. [07]

OR

- Q - 6 (a) Sketch the Root Locus for [07]

$$G(s)H(s) = \frac{K}{s(s+4)(s+2)}$$

- (b) A system is described in Figure 8, Determine its State equations. [07]

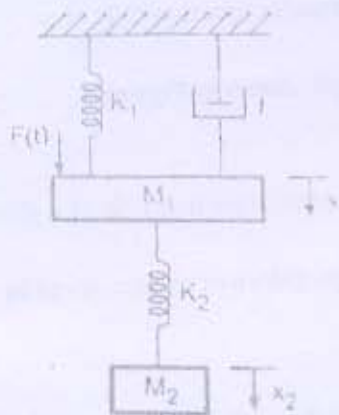


Figure 1.

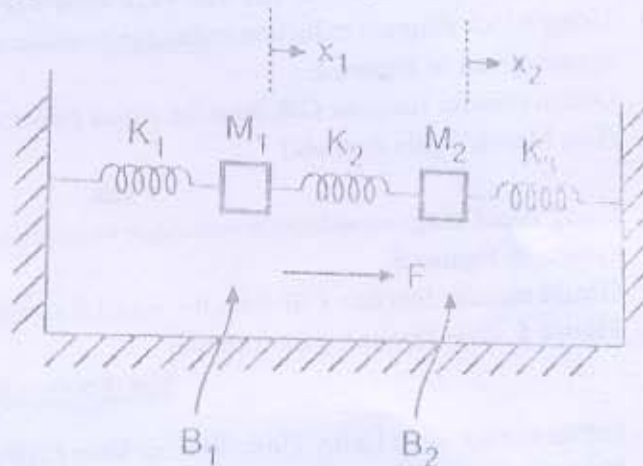


Figure 2

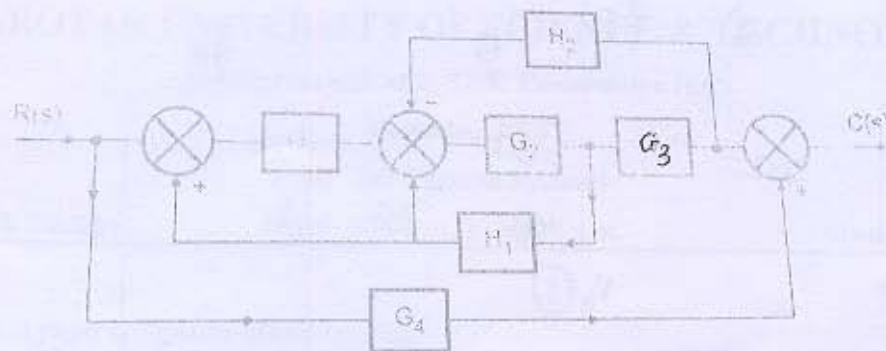


Figure 3

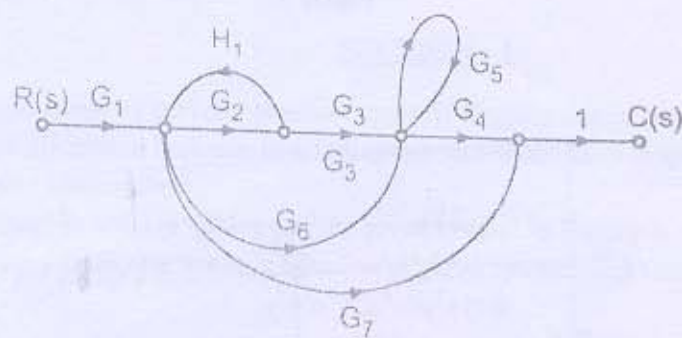


Figure 4

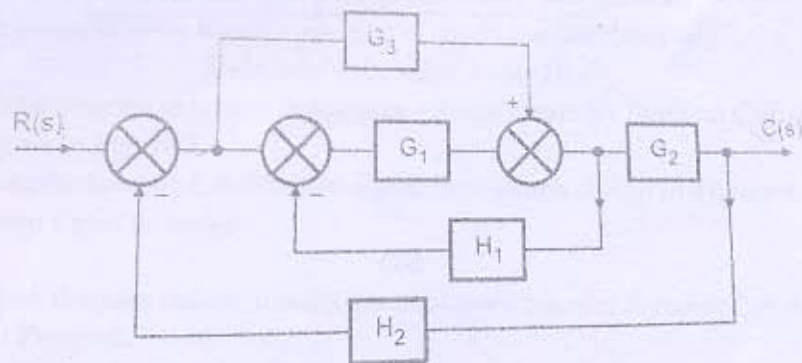


Figure 5

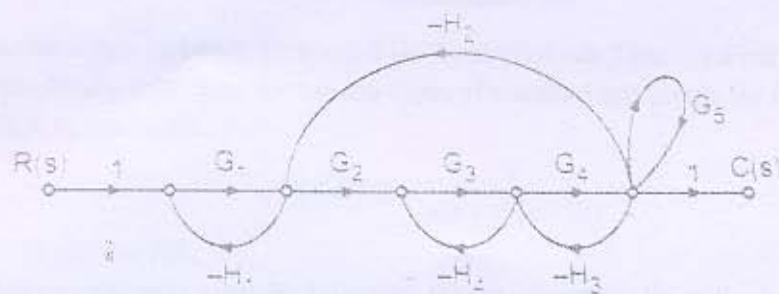


Figure 6

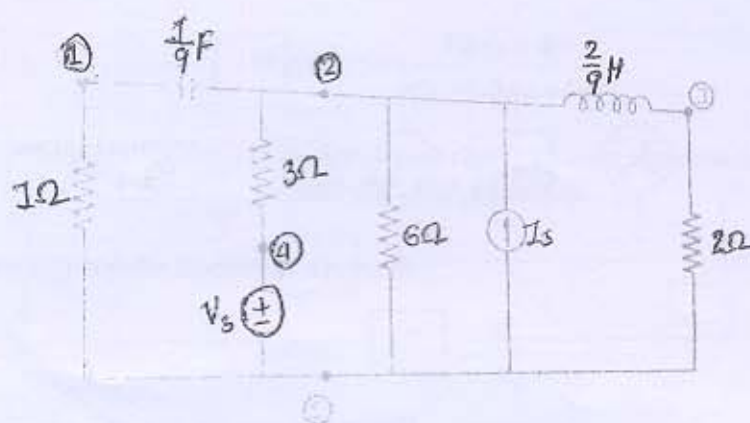


Figure 7.

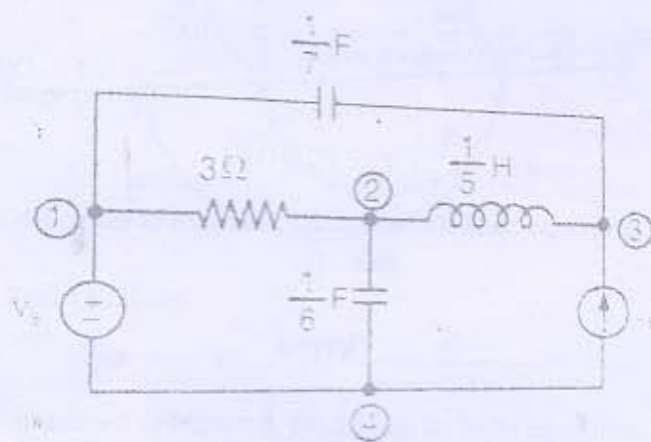


Figure 8.